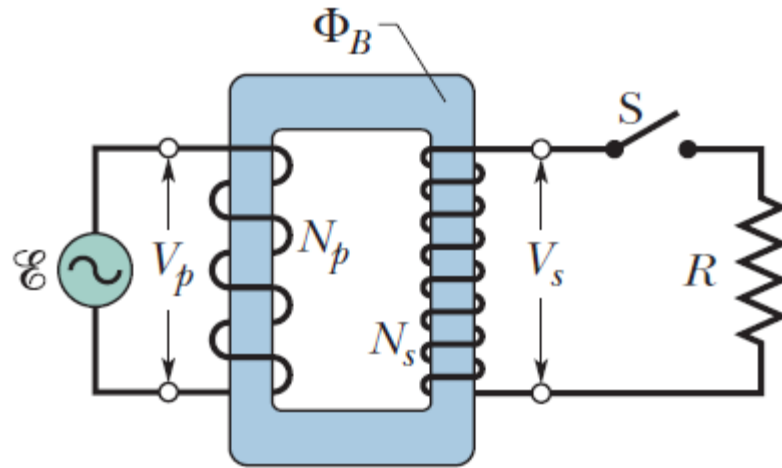
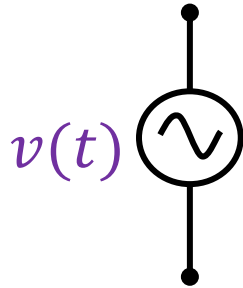


Alternating Current

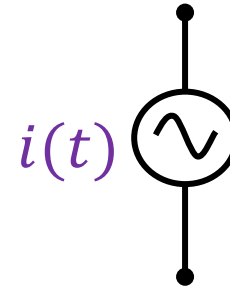


Phasors and Alternating Currents

AC Source:



or



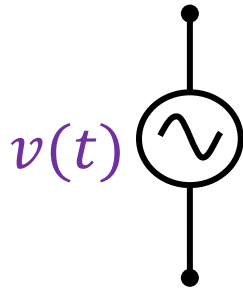
$$v(t) = V_0 \cos \omega t$$

$$i(t) = I_0 \cos (\omega t - \phi)$$

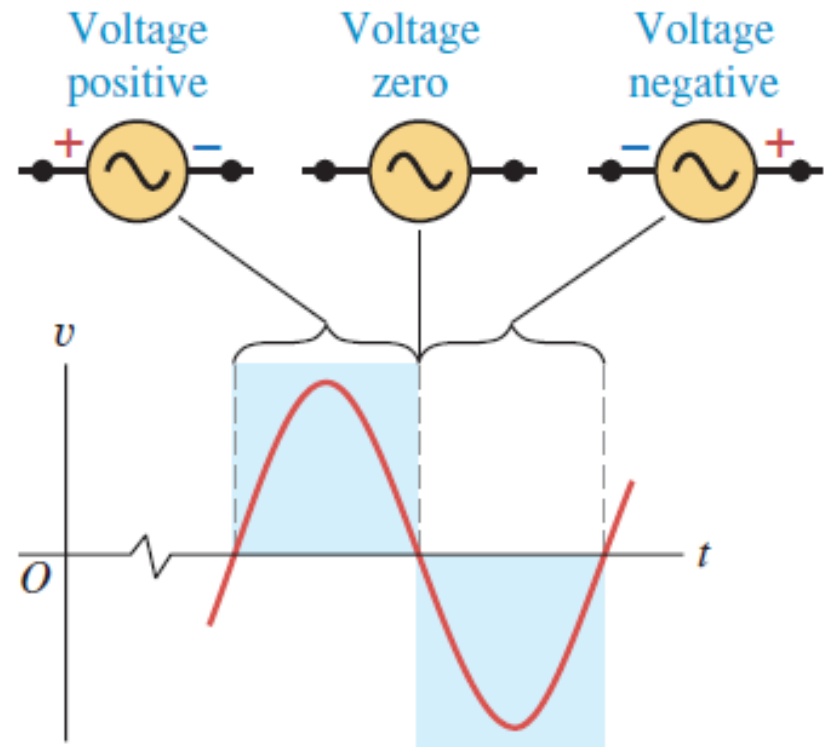
- V_0 and I_0 are the voltage and current amplitudes, respectively.
- ω is the angular frequency ($\omega = 2\pi f = 2\pi / T$).

Phasors and Alternating Currents

AC Source:



$$v(t) = V_0 \cos \omega t$$



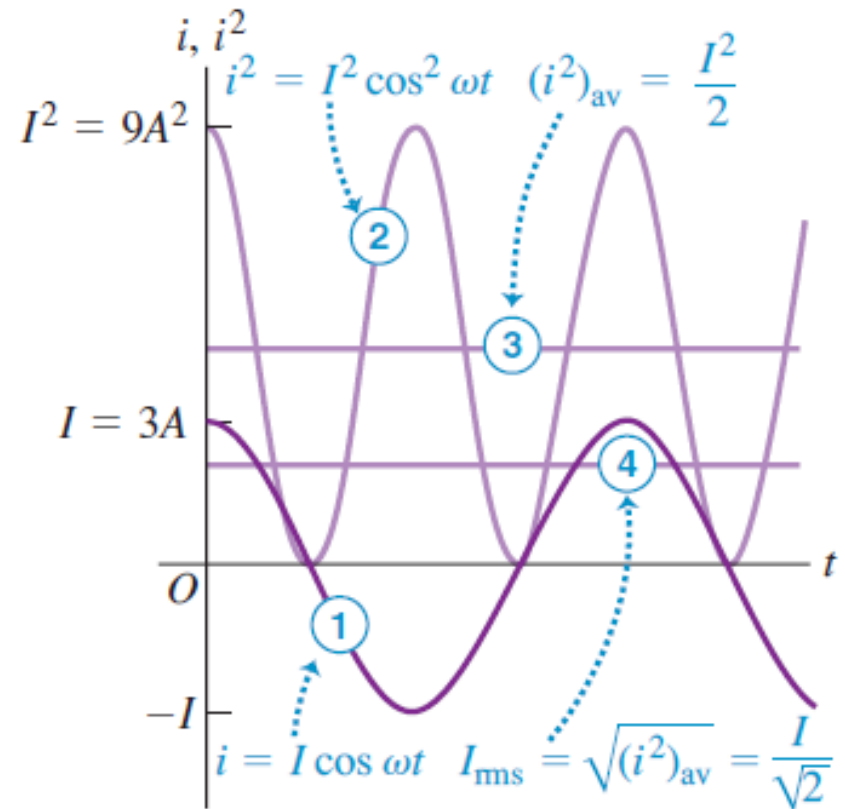
Phasors and Alternating Currents

AC Source:

- Root-mean-square value

$$i(t) = I_0 \cos \omega t$$

$$\begin{aligned} I_{rms} &= \sqrt{\frac{1}{T} \int_0^T i^2(t) dt} \\ &= \sqrt{\frac{1}{T} I_0^2 \int_0^T \cos^2(\omega t) dt} \\ &= \frac{I_0}{\sqrt{2}} \end{aligned}$$



Phasors and Alternating Currents

AC Source:

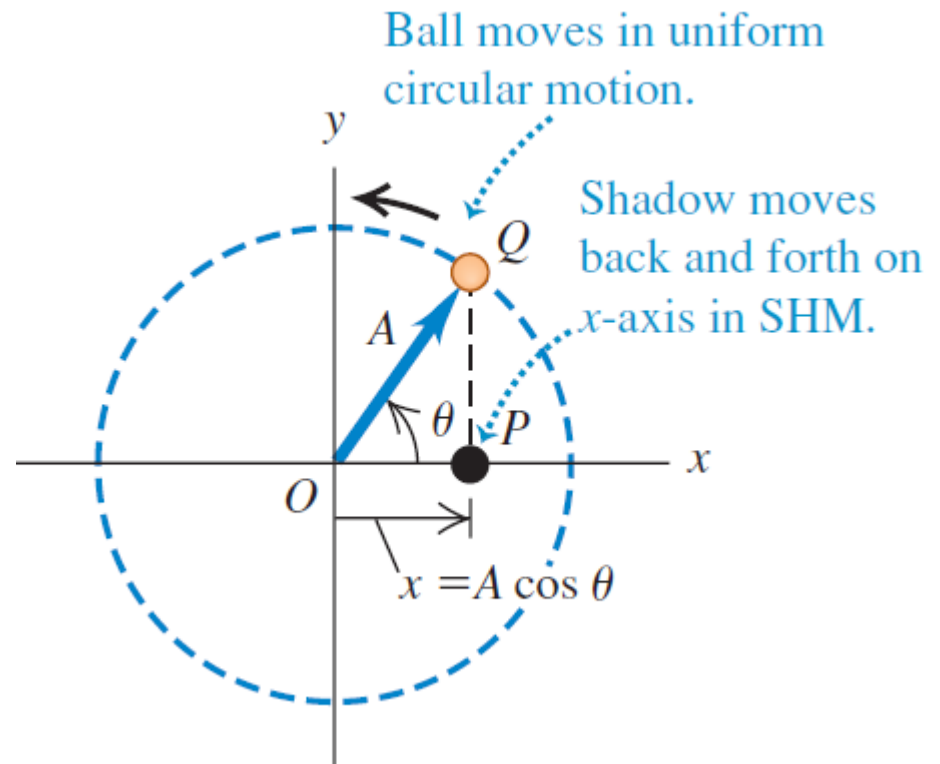
Phasor representation

- Let $\vec{V} = \vec{A}$

$$A_x = v = V_0 \cos \theta$$

- Let $\omega t = \theta$

$$v(t) = V_0 \cos \omega t$$



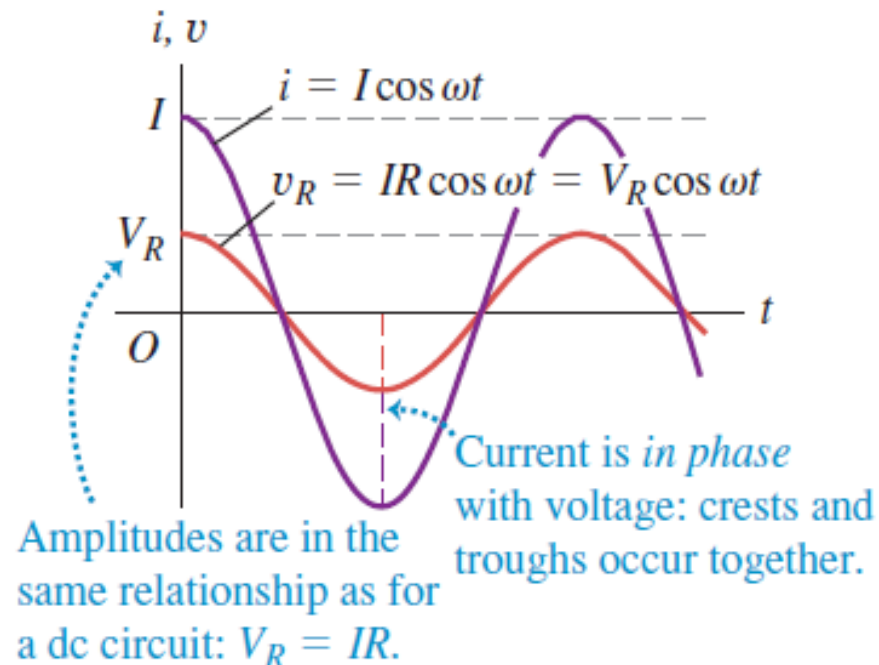
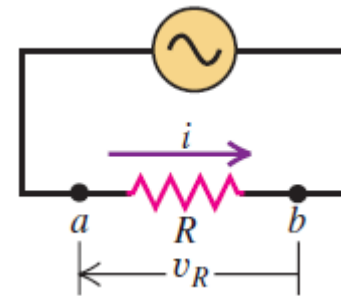
Resistance and Reactance

Resistance in an AC Circuit:

- Let $i = I \cos \omega t$

$$v_R = iR = IR \cos \omega t$$

- Time graph:



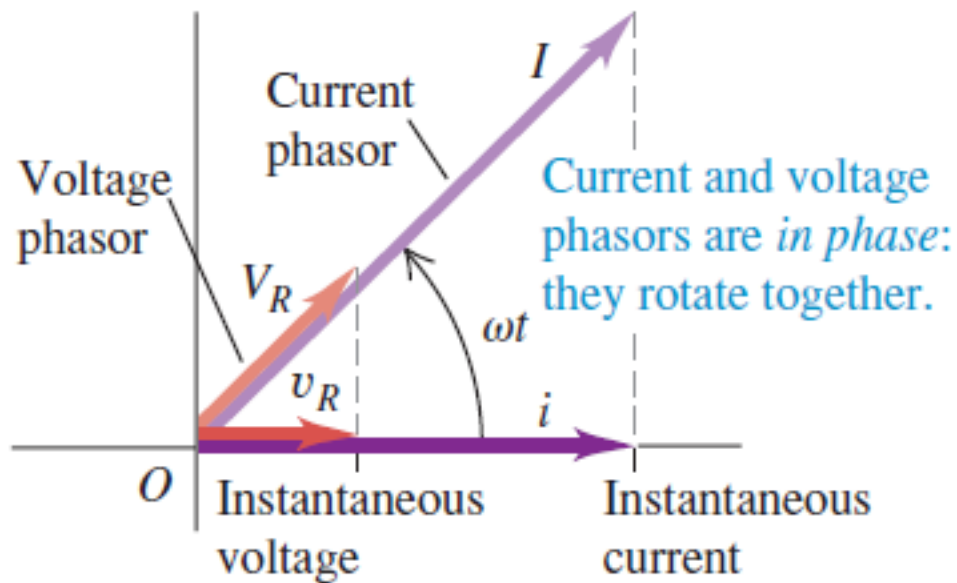
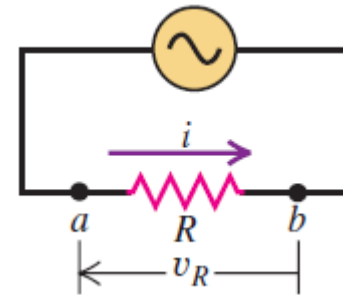
Resistance and Reactance

Resistance in an AC Circuit:

$$i = I \cos \omega t$$

$$v_R = IR \cos \omega t$$

- *Phasor diagram:*



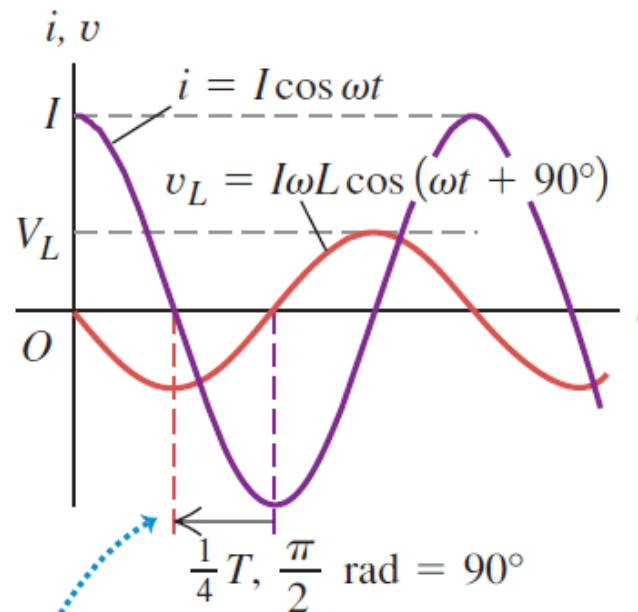
Resistance and Reactance

Inductance in an AC Circuit:

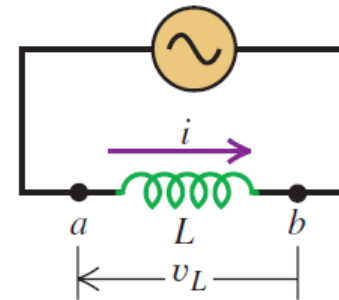
- Let $i = I \cos \omega t$

$$v_L = L \frac{di}{dt} = -I(\omega L) \sin \omega t$$

- Time graph:



Voltage curve *leads* current curve by a quarter-cycle (corresponding to $\phi = \pi/2 \text{ rad} = 90^\circ$).

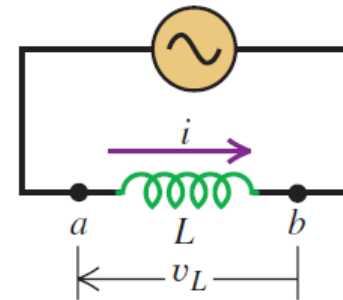


Resistance and Reactance

Inductance in an AC Circuit:

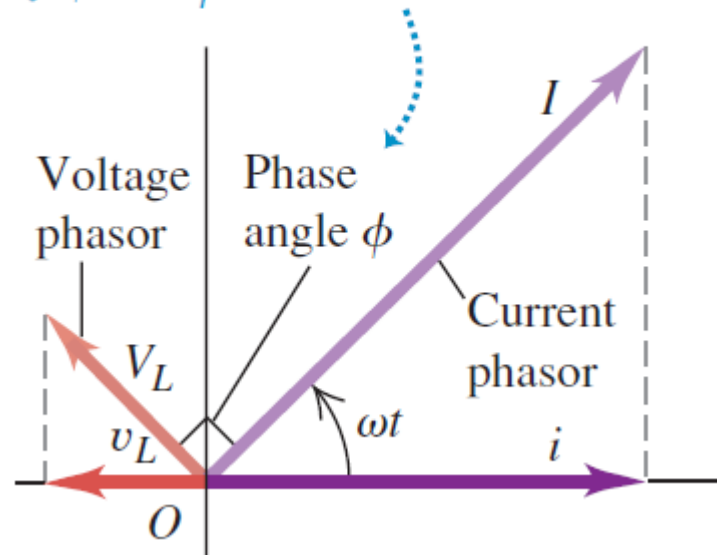
$$i = I \cos \omega t$$

$$\begin{aligned} v_L &= -I(\omega L)\sin \omega t \\ &= I(\omega L)\cos(\omega t + 90^\circ) \end{aligned}$$



- *Phasor diagram:*

Voltage phasor *leads* current phasor
by $\phi = \pi/2$ rad = 90° .



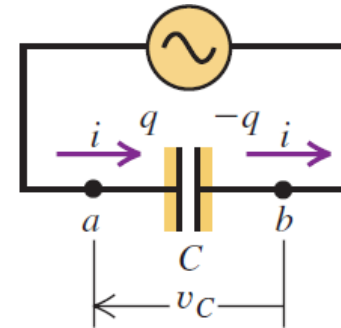
Resistance and Reactance

Capacitance in an AC Circuit:

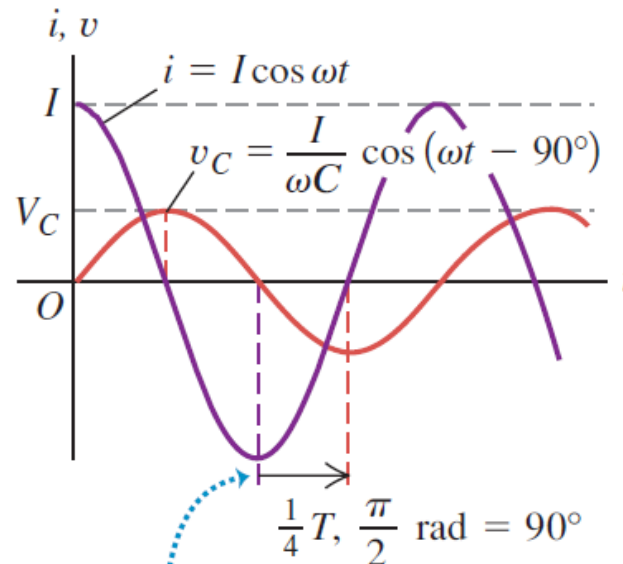
- Let $i = I \cos \omega t$

$$q = \int i \, dt = \frac{I}{\omega} \sin \omega t$$

$$v_C = \frac{q}{C} = \frac{I}{\omega C} \sin \omega t$$



- Time graph:



Voltage curve *lags* current curve by a quarter-cycle (corresponding to $\phi = -\pi/2 \text{ rad} = -90^\circ$)

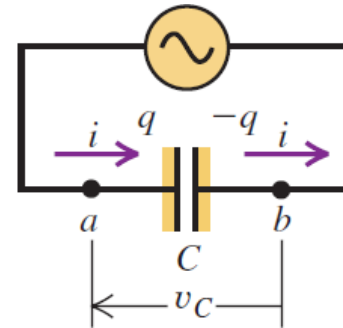
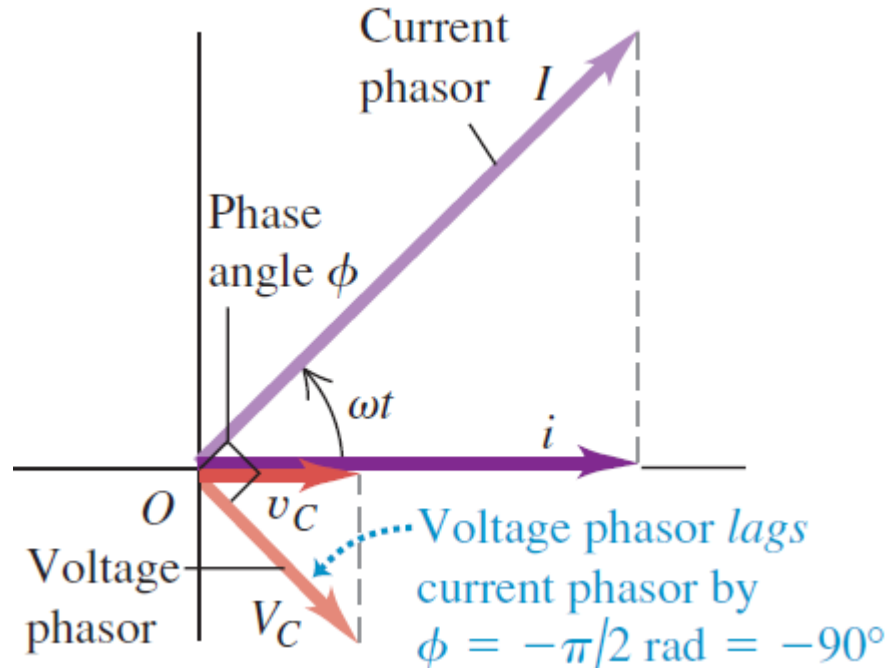
Resistance and Reactance

Capacitance in an AC Circuit:

$$i = I \cos \omega t$$

$$v_C = \frac{I}{\omega C} \sin \omega t = \frac{I}{\omega C} \cos(\omega t - 90^\circ)$$

- Phasor diagram:



Resistance and Reactance

Reactance:

$$v_R = I(R) \cos \omega t$$

$$v_L = I(\omega L) \cos(\omega t + 90^\circ) \quad v_C = \frac{I}{(\omega C)} \cos(\omega t - 90^\circ)$$

- *Inductive and capacitive reactance:*

$$v_L = I(X_L) \cos(\omega t + 90^\circ) \quad v_C = I(X_C) \cos(\omega t - 90^\circ)$$

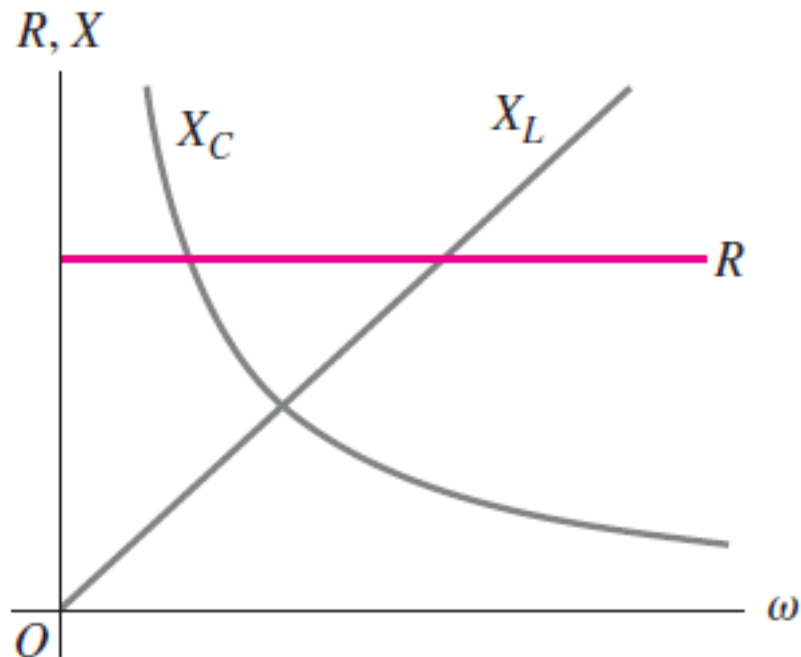
$$X_L = \omega L \quad X_C = \frac{1}{\omega C}$$

Think of these as frequency-dependent resistances!

Resistance and Reactance

Circuit elements in an AC Circuit:

Circuit Element	Amplitude Relationship	Circuit Quantity	Phase of v
Resistor	$V_R = IR$	R	In phase with i
Inductor	$V_L = IX_L$	$X_L = \omega L$	Leads i by 90°
Capacitor	$V_C = IX_C$	$X_C = 1/\omega C$	Lags i by 90°



The LRC Series Circuit

Series Circuit:

- Voltage rule:

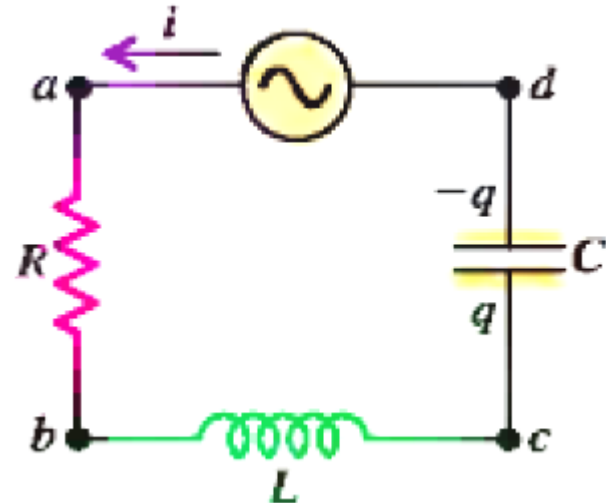
$$v_S = v_C + v_R + v_L$$

$$v_S = \frac{q}{C} + Ri + L \frac{di}{dt}$$

- As $i = dq/dt$

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = v_S(t) \quad \text{or} \quad L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{i}{C} = v'_S(t)$$

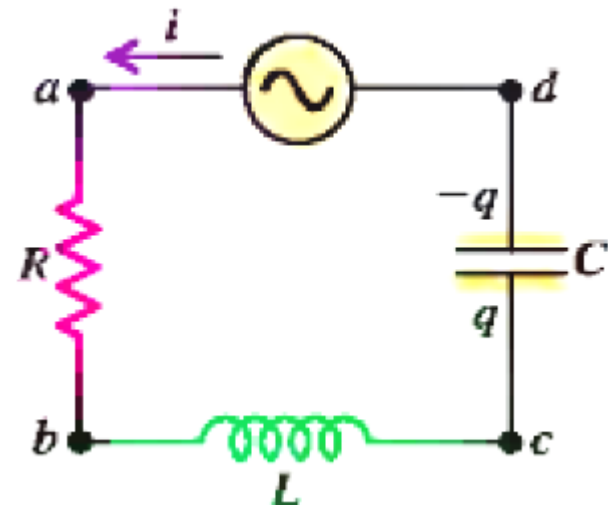
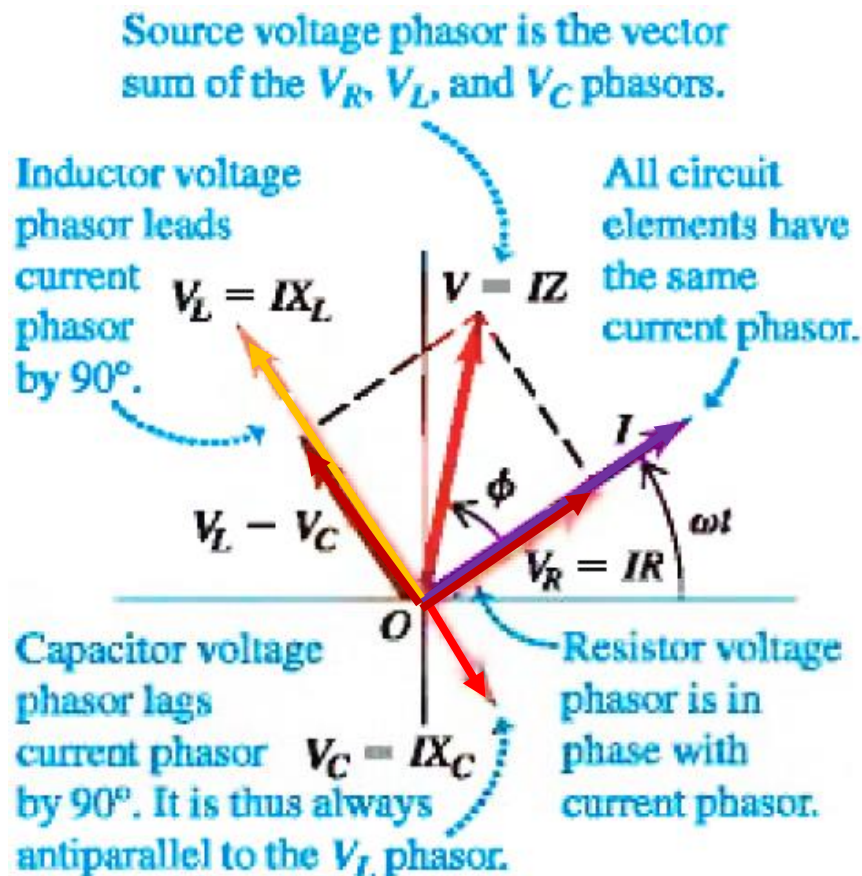
2nd order differential equation!



The LRC Series Circuit

Series Circuit:

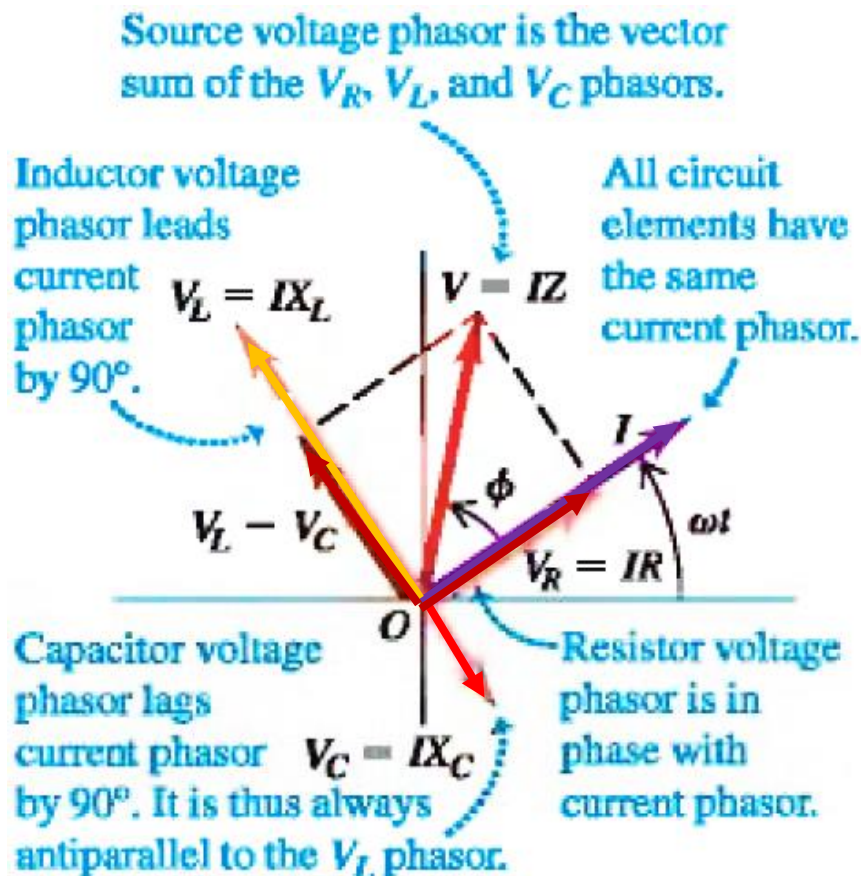
- Using phasor method:



The LRC Series Circuit

Series Circuit:

- Using phasor method:



$$\vec{V} = \vec{V}_R + \vec{V}_L + \vec{V}_C$$

$$V^2 = (IR)^2 + (IX_L - IX_C)^2$$

$$\frac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\tan \phi = \frac{(X_L - X_C)}{R}$$

The LRC Series Circuit

Series Circuit:

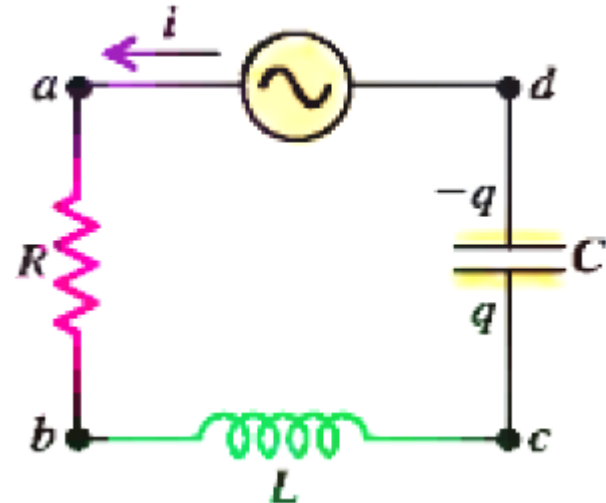
- Using phasor method:

$$\frac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2} \equiv Z$$

$$\tan \phi = \frac{(X_L - X_C)}{R}$$

$$i(t) = I \cos(\omega t)$$

$$v(t) = IZ \cos(\omega t + \phi)$$



The LRC Series Circuit

Example: Series circuit

Suppose $R = 300 \, \Omega$, $L = 60 \, \text{mH}$, $C = 0.50 \, \mu\text{F}$, $V = 50 \, \text{V}$, and $\omega = 10,000 \, \text{rad/s}$. Find X_L , X_C , Z , I , ϕ and the voltage amplitude across each element.

Solution:

- Reactance and impedance:*

$$X_L = \omega L = (10,000 \, \text{s}^{-1})(60 \times 10^{-3} \, \text{H}) = 600 \, \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{(10,000 \, \text{s}^{-1})(0.5 \times 10^{-6} \, \text{F})} = 200 \, \Omega$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{(300)^2 + (600 - 200)^2} = 500 \, \Omega$$

The LRC Series Circuit

Example: Series circuit

Solution:

- Phase angle:*

$$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right) = \tan^{-1} \left(\frac{400}{300} \right) = 53^\circ$$

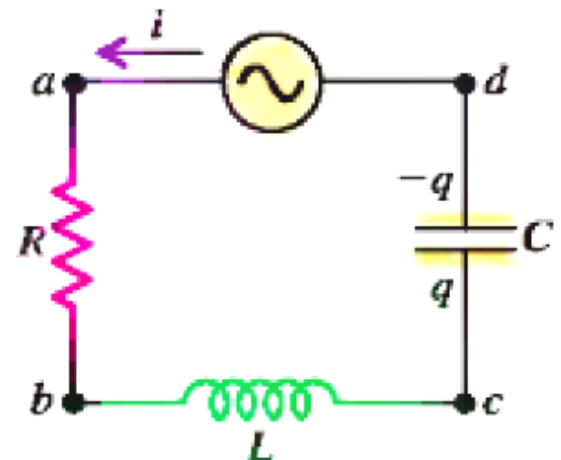
- Current and voltage amplitudes:*

$$I = \frac{V}{Z} = \frac{50 \text{ V}}{500 \Omega} = 0.10 \text{ A}$$

$$V_R = IR = (0.1 \text{ A})(300 \Omega) = 30 \text{ V}$$

$$V_L = IX_L = (0.1 \text{ A})(600 \Omega) = 60 \text{ V}$$

$$V_C = IX_C = (0.1 \text{ A})(200 \Omega) = 20 \text{ V}$$



The LRC Series Circuit

Example: Series circuit

Solution:

- The current and voltages:* $\omega = 10,000$ rad/s

$$i(t) = I \cos(\omega t) = (0.10)\cos(\omega t)$$

$$v_R(t) = V_R \cos(\omega t) = (30)\cos(\omega t)$$

$$v_L(t) = -V_L \sin(\omega t) = -(60)\sin(\omega t)$$

$$v_C(t) = V_C \sin(\omega t) = (20)\sin(\omega t)$$

$$v(t) = IZ \cos(\omega t + \phi) = (50)\cos(\omega t + 53^\circ)$$

The LRC Series Circuit

Example: Series circuit

Solution (final):

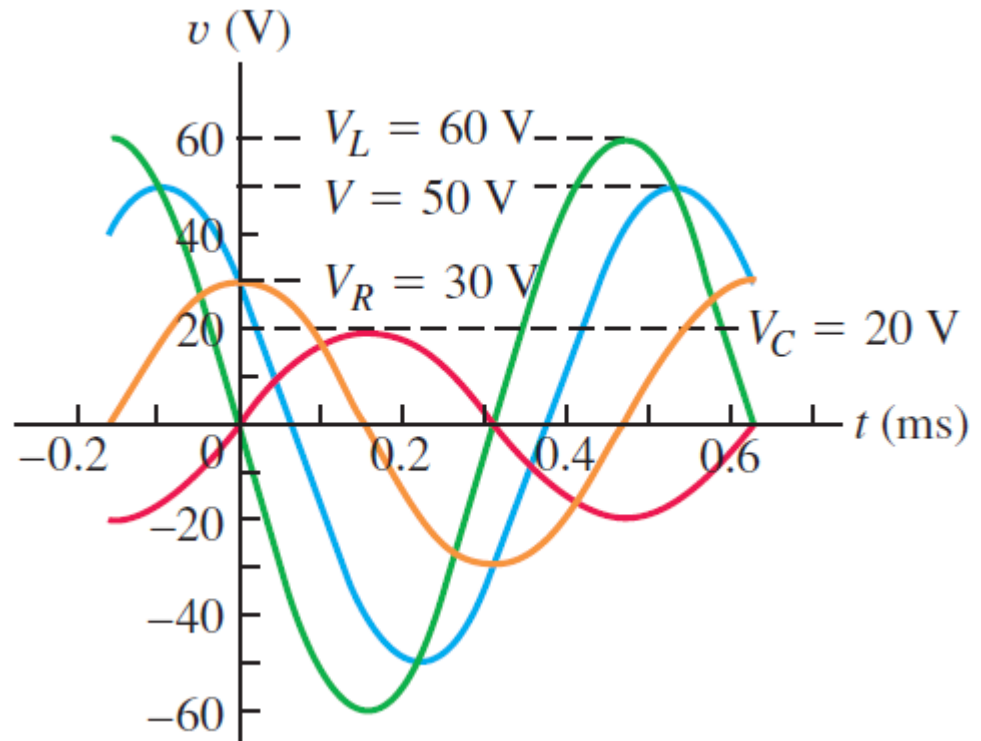
$$i(t) = (0.10)\cos(\omega t)$$

$$v_R(t) = (30)\cos(\omega t)$$

$$v_L(t) = -(60)\sin(\omega t)$$

$$v_C(t) = (20)\sin(\omega t)$$

$$v(t) = (50)\cos(\omega t + 53^\circ)$$



Power in AC Circuits

Power: General case

$$i(t) = I \cos \omega t \qquad v(t) = V \cos(\omega t + \phi)$$

- *Instantaneous power:*

$$\begin{aligned} P(t) &= v i = [V \cos(\omega t + \phi)][I \cos \omega t] \\ &= VI[(\cos \omega t \cos \phi - \sin \omega t \sin \phi)][\cos \omega t] \\ &= VI[\cos^2 \omega t \cos \phi - \sin \omega t \cos \omega t \sin \phi] \end{aligned}$$

- *Average power:*

$$\begin{aligned} P_{\text{ave}} &= (v i)_{\text{ave}} \\ &= VI\left[\left(\frac{1}{2}\right)\cos \phi - 0\right] \\ &= \frac{1}{2} VI(\cos \phi) = V_{\text{rms}} I_{\text{rms}} \cos \phi \end{aligned}$$

Power in AC Circuits

Power: General case

- Average power:*

$$P_{\text{ave}} = V_{\text{rms}} I_{\text{rms}} \cos \phi$$

$$\cos \phi = \text{Power factor}$$

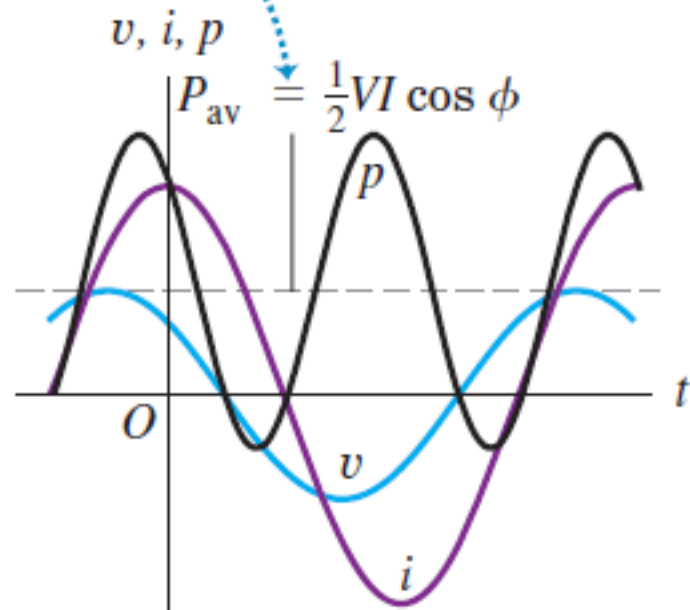
- Recall:*

$$\tan \phi = \frac{(X_L - X_C)}{R}$$

$$\cos \phi = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$$

- For purely resistive circuit: $\cos \phi = 1$
- For purely inductive (capacitive) circuit: $\cos \phi = 0$

For an arbitrary combination of resistors, inductors, and capacitors, the average power is positive.

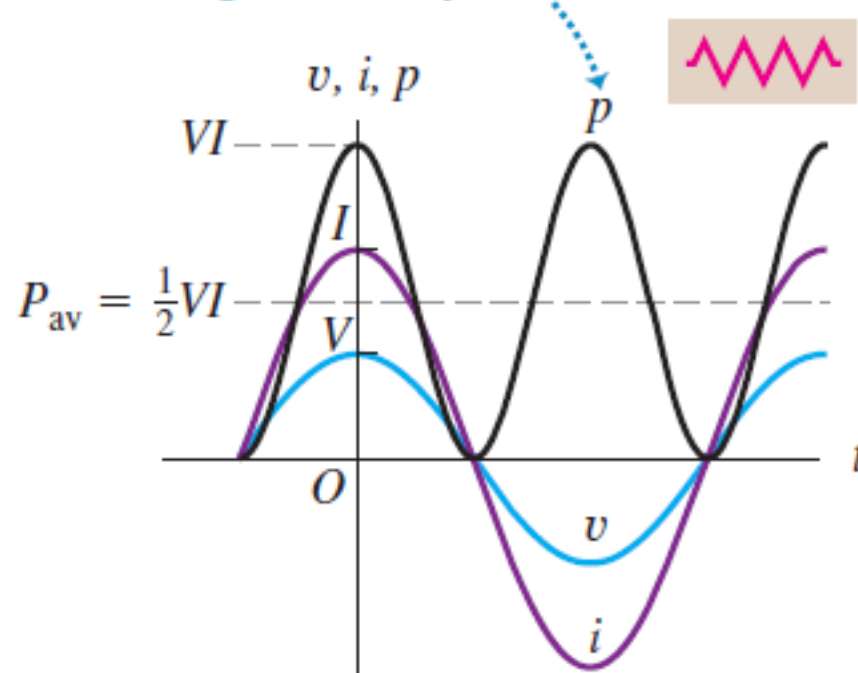


Power in AC Circuits

Power:

- For purely resistive circuit: $\cos \phi = 1$

For a resistor, $p = vi$ is always positive because v and i are either both positive or both negative at any instant.

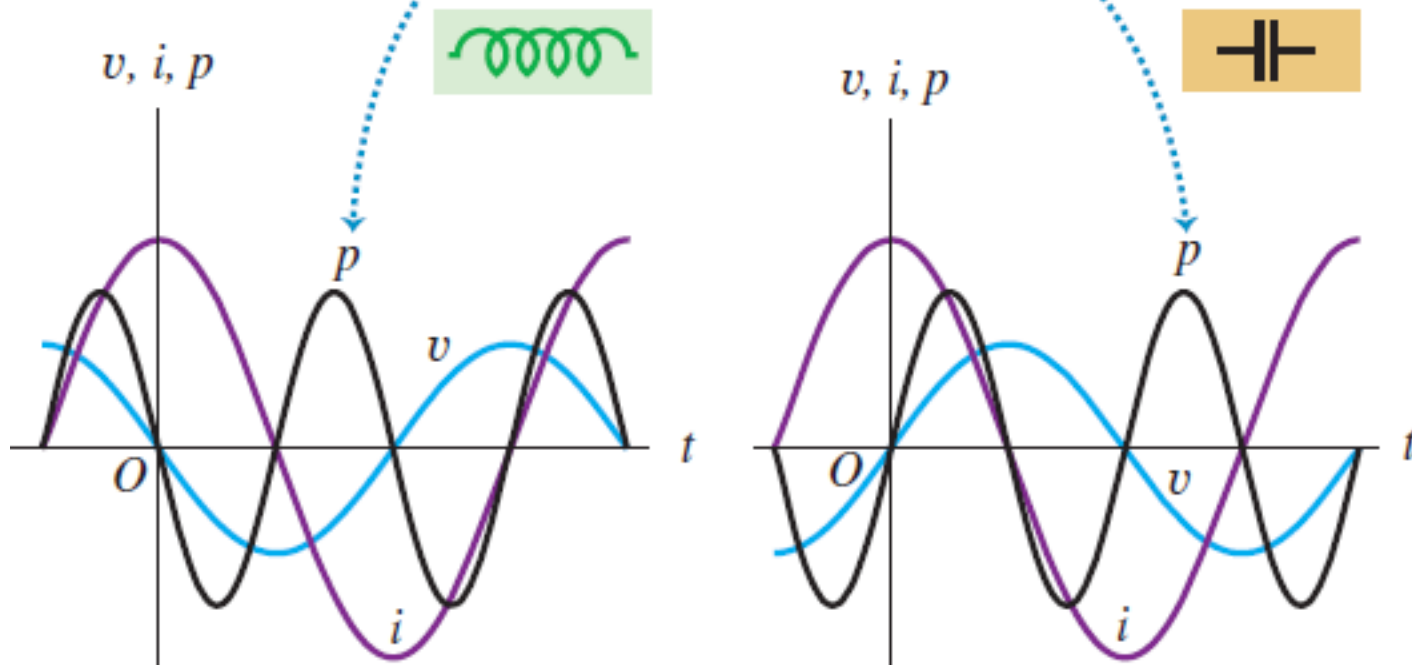


Power in AC Circuits

Power:

- For purely inductive (capacitive) circuit: $\cos \phi = 0$

For an inductor or capacitor, $p = vi$ is alternately positive and negative, and the average power is zero.



Resonance in AC Circuits

Impedance and frequency:

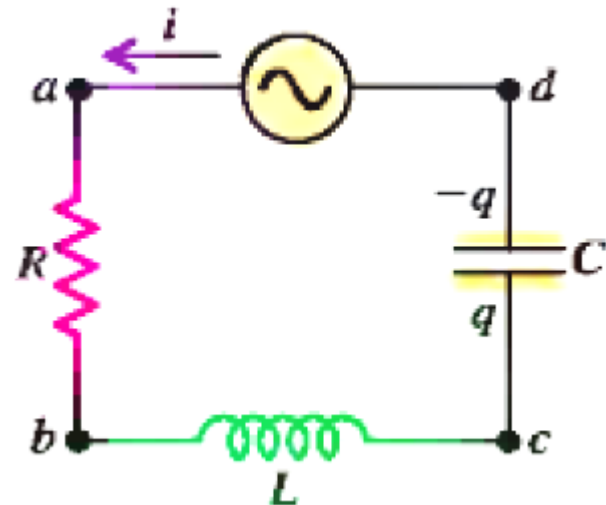
$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$
$$= \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

- *Minimum impedance:*

$$Z = R \quad \text{when} \quad X_L = X_C$$

- *Resonance:*

$$\omega L = \frac{1}{\omega C} \quad \omega = \frac{1}{\sqrt{LC}} = \omega_0$$

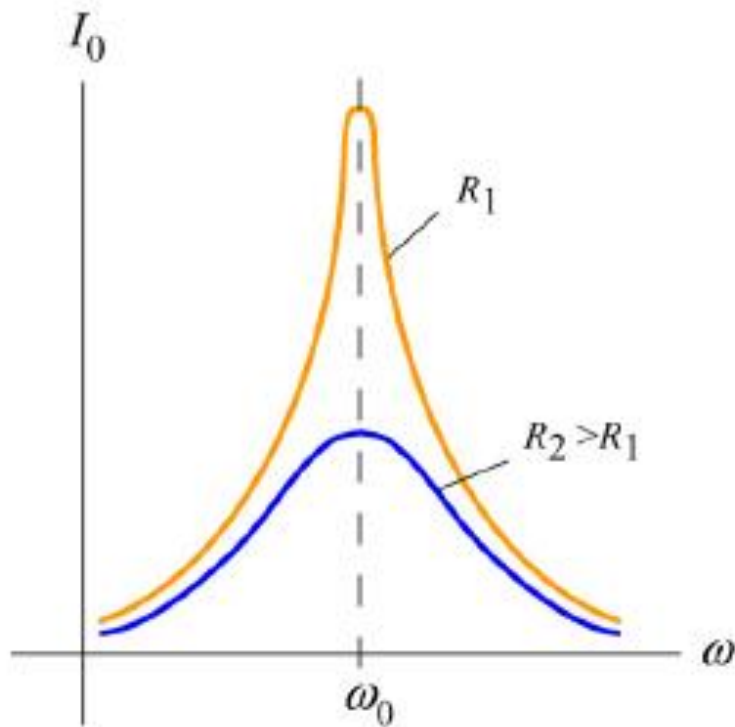


Resonance in AC Circuits

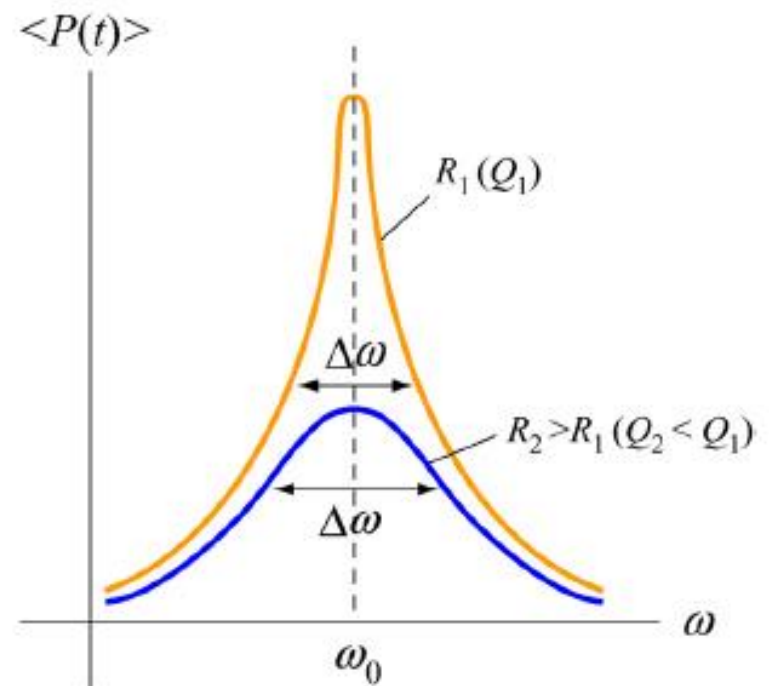
Impedance and frequency:

- *Resonance:*

$$\omega = \omega_0$$



Current amplitude vs. ω



Average power vs. ω

Resonance in AC Circuits

Example: Resonance in a series circuit

A series circuit similar to some radio tuning circuit has $R = 500 \, \Omega$, $L = 0.40 \, \text{mH}$, $C = 100 \, \text{pF}$. It is connected to a variable-frequency ac source with an RMS terminal voltage of $1.0 \, \text{V}$. (a) Find the resonance frequency.

At the resonance frequency find (b) X_L , X_C and Z ; and (c) the RMS current and the voltage amplitude across each element.

Solution:

a) Resonance frequency:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.4 \times 10^{-3} \, \text{H})(100 \times 10^{-12} \, \text{F})}} = 5 \times 10^6 \, \text{rad/s}$$

$$f_0 = 8.0 \times 10^5 \, \text{Hz} = 800 \, \text{kHz}$$

Resonance in AC Circuits

Example: Resonance in a series circuit

Solution:

b) Reactance and impedance:

$$X_C = \frac{1}{\omega C} = \frac{1}{(5 \times 10^6 \text{ rad/s})(100 \times 10^{-12} \text{ F})} = 2000 \Omega$$

$$X_L = \omega L = (5 \times 10^6 \text{ rad/s})(0.4 \times 10^{-3} \text{ H}) = 2000 \Omega$$

$$Z = R = 500 \Omega$$

c) RMS voltages:

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{1 \text{ V}}{500 \Omega} = 0.002 \text{ A}$$

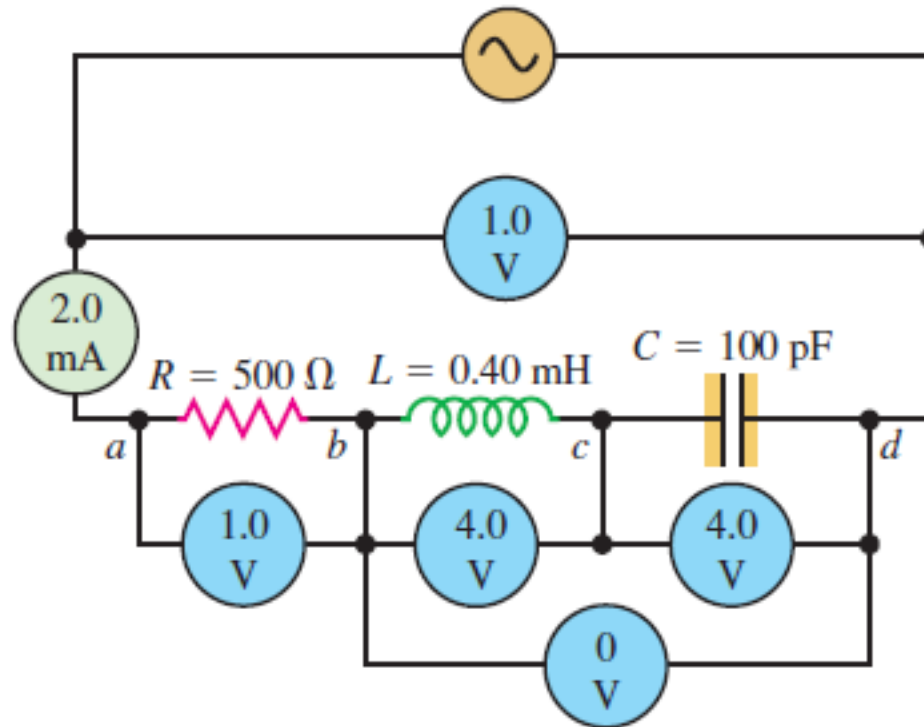
Resonance in AC Circuits

Example: Resonance in a series circuit

Solution:

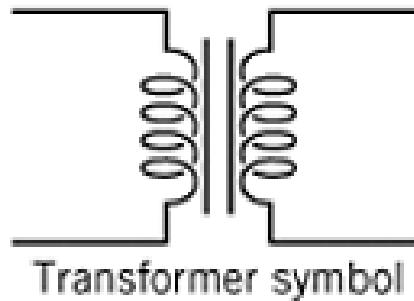
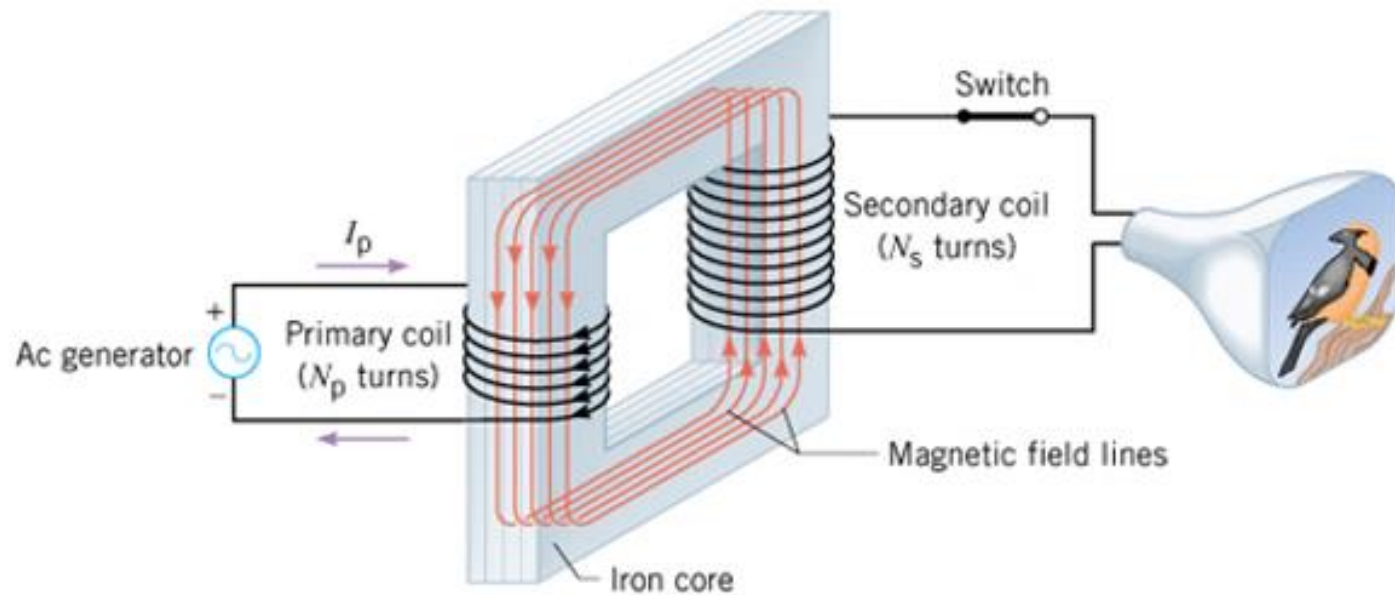
c) Rms voltages:

$$I_{rms} = 2.0 \text{ mA}$$



Transformers

Ideal transformer:



Transformers

Ideal transformer:

- In primary coil:*

$$V_1 = -N_1 \frac{d\Phi_m}{dt}$$

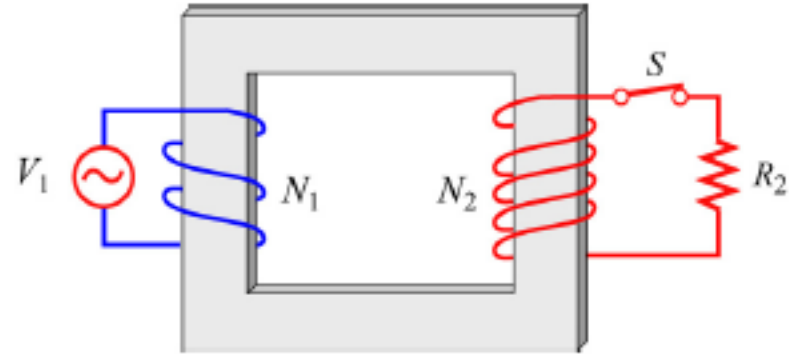
- In secondary coil:*

$$V_2 = -N_2 \frac{d\Phi_m}{dt}$$

- With complete flux linkage:*

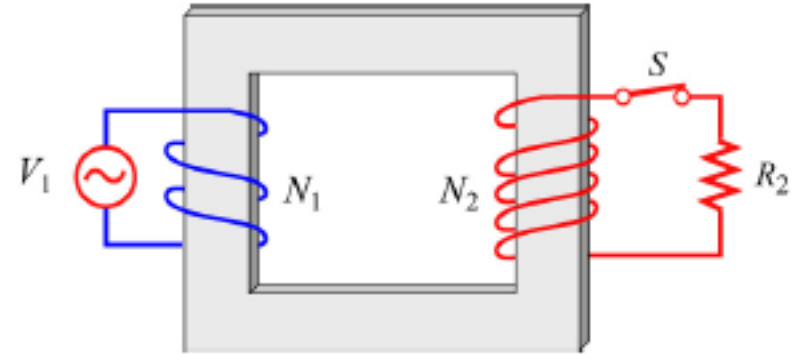
$$\frac{V_2}{N_2} = \frac{V_1}{N_1} \Rightarrow \frac{V_2}{V_1} = \frac{N_2}{N_1}$$

*Transformer
equation!*



Transformers

Ideal transformer:



- *With no power losses:*

$$V_1 I_1 = V_2 I_2 \quad \Rightarrow \quad I_1 = \left(\frac{N_2}{N_1} \right) I_2$$

- *Combining:*

$$\frac{V_1}{I_1} = \left(\frac{N_1}{N_2} \right)^2 R_2$$

The transformer “transforms” voltage, currents, as well as resistance!

Transformers

Example: Imported coffeemaker

A friend returns to the United States from Europe with a 960-W coffeemaker, designed to operate from a 240-V line. (a) What can she do to operate it at the USA-standard 120 V? (b) What current will the coffeemaker draw from the 120-V line? (c) What is the resistance of the coffeemaker? (The voltages are RMS values.)

Solution:

a) Use step-up transformer:

$$\frac{N_2}{N_1} = \frac{V_2}{V_1} = \frac{240 \text{ V}}{120 \text{ V}} = 2$$

Transformers

Example: Imported coffeemaker

Solution:

b) From average power $P_{ave} = V_1 I_1$:

$$I_1 = \frac{960 \text{ W}}{120 \text{ V}} = 8 \text{ A} \Rightarrow I_2 = \frac{960 \text{ W}}{240 \text{ V}} = 4 \text{ A}$$

c) Resistance:

$$R_2 = \left(\frac{N_2}{N_1} \right)^2 \frac{V_1}{I_1} = (2)^2 (15 \Omega) = 60 \Omega$$

END

